Patuakhali Science and Technology University

B.Sc.Engg.(CSE) Level-1 Semester-I Mid Term Examination-2018 (January-June)

Course Code: EEE 111 Course Title: Basic Electrical Engineering

Credit Hour: 3.0 Full Marks: 15 Duration: 1 Hour.

1. State and explain Ohm's law. Apply Kirchhoff's voltage law, to find the values of current i and the voltage drops V_1 and V_2 in the circuit shown in fig: 1 — 3

2. Find the node voltage drops V_1 and V_2 using the following methods for the fig. 2 — 3*4=12
   a. Mesh Analysis
   b. Nodal Analysis
   c. Superposition Method

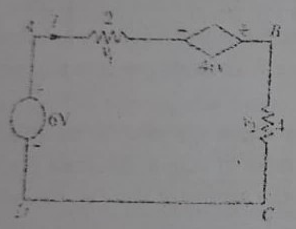


Fig: 1

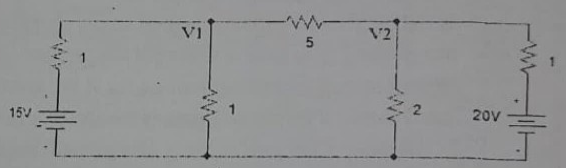


Fig. 2

Patuakhali Science and TechnologyUniversity

B.Sc.Engg.(CSE) Level-1 Semester-I Final Examination-2018 (January-June)

Course Code: EEE 111 Course Title: Basic Electrical Engineering

Credit Hour: 3.0 Full Marks: 70 Duration: 3 Hours.

[Figures in the right margin indicate full marks. Split answering of any question is not recommended.]

Answer any 5 of the following questions.

All values of resistances are in Ohm

(a) Define circuit. Explain short circuits and open circuits. 04

(b) State and explain the basic laws of electrical engineering. 04

(c) What are the laws of resistance? A 50-ohm resistor is in parallel with 100 ohm resistor. Current in 50 ohm resistor is 7.2 A. How will you add a third resistor and what will be its value of the line current if line current is to be 12.1 A? 03

(d) Write down the main properties of series and parallel circuit. 03

(a) Determine the value of current through 5 Ω resistor as shown in fig: 2(a) 04

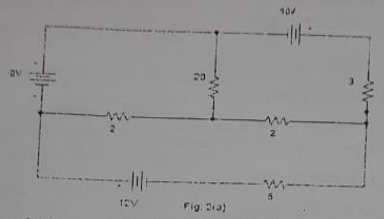


Fig: 2(a)

(b) Use nodal analysis to find i_0 current in the circuit of fig: 2(b) 05

(c) Use mesh analysis to find i_0 current in the circuit of fig: 2(c) 05

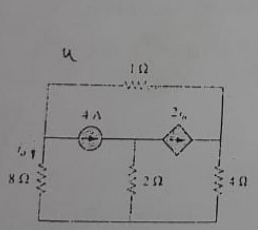


fig: 2(b)

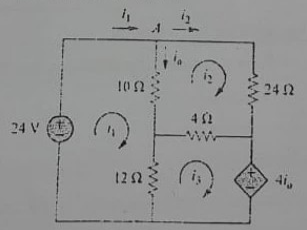


fig: 2(c)

3 (a) Define magnetic circuit. Make a comparison between magnetic and electric circuit. 03

(b) Find current through a load of 2 Ω in the circuit shown in fig:3(b) 05

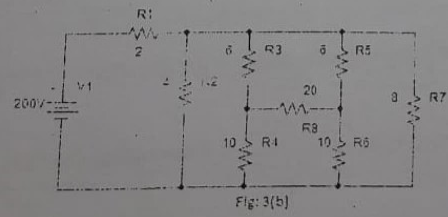


Fig: 3(b)

a). "Magnetic circuit is analogous to an electrical circuit"- Justify this statement with relevant diagram.

b). Find current using Thevenin's theorem through a load of 8 Ω in the circuit shown in fig.4(b) 5

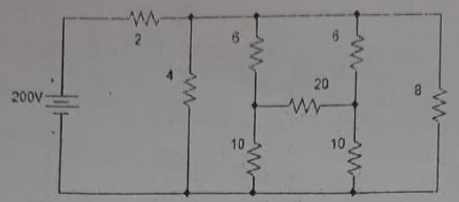


fig. 4(b)

c). State and explain Fleming's right hand rule for induced current. 2

d). A circuit having a resistance of 5Ω and inductive reactance of 8Ω is connected in parallel with another circuit having a resistance of 8Ω and capacitive reactance of 5Ω. The parallel circuit is connected across 200V, 50 Hz supply. Calculate: (i) supply current (ii) power factor of the whole circuit (iii) power consumed and (iv) the resistance and reactance of a series circuit which will take the same current at the same p.f. as the parallel circuit. 5

3 a). Drive the equations of alternating current and voltage of an a.c circuit containing capacitance and series resistance. 4

b). Define effective value of an alternating current. Determine R.M.S value and average value of the half sine waveform in fig 5(b). 6

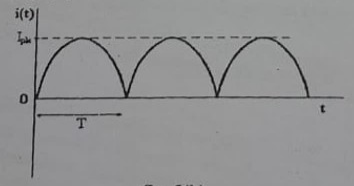


fig. 5(b)

c). A pure inductive coil allows a current of 12A to flow from a 240V, 50 Hz supply. Find (i) inductive reactance (ii) inductance of the coil (iii) power absorbed. Write down the equations for voltage and current. 4

4 a). Define imaginary value (j), power factor and phasor diagram. 3

b). Define resonance in R-L-C series circuit. What are the effects of series resonance? Discuss the bandwidth of a series resonance circuit. 5

c). The voltage applied to a parallel circuit is 58+j53 volts. The current in one branch is 5- j7 amperes and the current in the other branch is 3+j7 amperes. What is the circuit power factor? 2

d). A 150 V, 50W lamp is to be operated on 220 V, 50 Hz supply. Find the value of (i) non-inductive resistance (ii) pure inductance in series with the lamp so that the lamp is not overrun. Which would be preferable? 4

Patuakhali Science and Technology University

B.Sc.Engg.(CSE) Level-1 Semester-I Final Examination-2017 (January-June)

Course Code: EEE 111 Course Title: Basic Electrical Engineering

Credit Hour: 3.0 Full Marks: 70 Duration: 3 Hours.

[Figures in the right margin indicate full marks. Split answering of any question is not recommended.]

Answer any 5 of the following questions.

1 a). Discuss open circuit and short circuit. What are the basic laws of electrical circuits? Mathematically state them. 4

b). An electric heater absorbs 1000W from mains 120V mains. By how much should its resistance be increased to reduce the power to 900W? 2

c). Four Bulbs of 25W, 40W, 60W, 100W each rated 250V are connected in series to supply main 220V. Which bulb will glow more and why? Show mathematically. 3

d). Calculate the equivalent resistance and the source current of the following circuit. 5

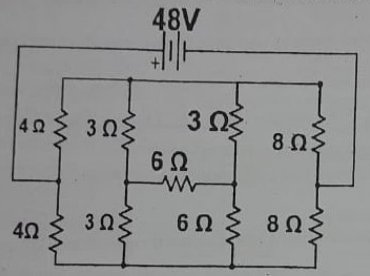


Fig.1(d)

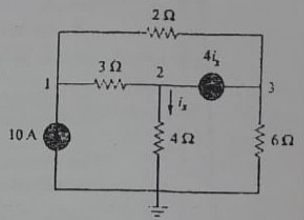


Fig.2(c)

2 a). State and explain reciprocity theorem. 3

b). Write down the properties of series and parallel circuit. 3

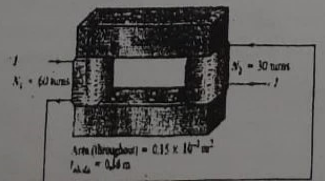
c). Find the voltages at the three nonreference nodes in the circuit of Fig.2(c) 4

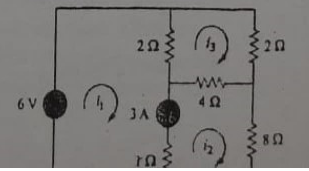
d). Define magnetic hysteresis. Draw and describe hysteresis loop of a magnetic material 4

3 a). For the series magnetic circuit of Fig.3(a) with two impressed sources of magnetic "pressure," determine the current I. Each applied mmf establishes a flux pattern in the clockwise direction. The value of Magnetizing Force is 20At/m. 4

b). With proper diagram deduce the condition for maximum power transmission. Also, write down the equation for maximum power and efficiency during this condition. 4

c). Use mesh analysis to determine i1, i2, and i3 in Fig. 3(c) 5





a). "Magnetic circuit is analogous to an electrical circuit"- Justify this statement with relevant diagram.

b). Find current using Thevenin's theorem through a load of 8 Ω in the circuit shown in fig.4(b) 5

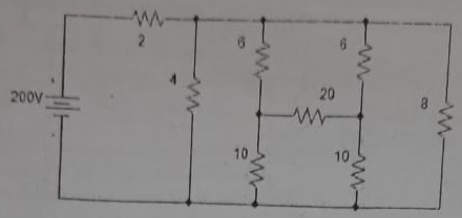


fig. 4(b)

c). State and explain Fleming's right hand rule for induced current. 2

d). A circuit having a resistance of 5Ω and inductive reactance of 8Ω is connected in parallel with another circuit having a resistance of 8Ω and capacitive reactance of 5Ω. The parallel circuit is connected across 200V, 50 Hz supply. Calculate: (i) supply current (ii) power factor of the whole circuit (iii) power consumed and (iv) the resistance and reactance of a series circuit which will take the same current at the same p.f. as the parallel circuit. 5

a). Drive the equations of alternating current and voltage of an a.c circuit containing capacitance and series resistance. 4

b). Define effective value of an alternating current. Determine R.M.S value and average value of the half sine waveform in fig 5(b). 6

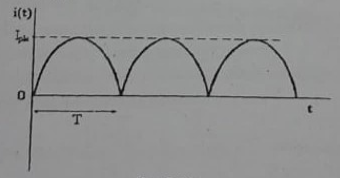


fig. 5(b)

c). A pure inductive coil allows a current of 12A to flow from a 240V, 50 Hz supply. Find (i) inductive reactance (ii) inductance of the coil (iii) power absorbed. Write down the equations for voltage and current. 4

a). Define imaginary value (j), power factor and phasor diagram. 3

b). Define resonance in R-L-C series circuit. What are the effects of series resonance? Discuss the bandwidth of a series resonance circuit. 5

c). The voltage applied to a parallel circuit is 58+j53 volts. The current in one branch is 5- j7 amperes and the current in the other branch is 3+j7 amperes. What is the circuit power factor? 2

d). A 150 V, 50W lamp is to be operated on 220 V, 50 Hz supply. Find the value of (i) non-inductive resistance (ii) pure inductance in series with the lamp so that the lamp is not overrun. Which would be preferable? 4

Patuakhali Science and Technology University

B.Sc Engg.(CSE) Level-1 Semester-I Final Examination-2016 (January-June)

Course Code: EEE 111 Course Title: Basic Electrical Engineering

Credit Hour: 3.0 Full Marks: 70 Duration: 3 Hours.

[Figures in the right margin indicate full marks. Split answering of any question is not recommended.]

Answer any 5 of the following questions.

All values of resistances are in Ohms

(a) Explain short circuits and open circuits. 03

(b) A resistance of 10 Ω is connected in series with two resistances each of 15 Ω arranged in parallel. What resistance must be shunted across this parallel combination so that the total current taken shall be 1.5 A with 20 V applied? 04

(c) A network arranged as shown in fig: 1(c). Determine the current in each resistance using loop current method. 04

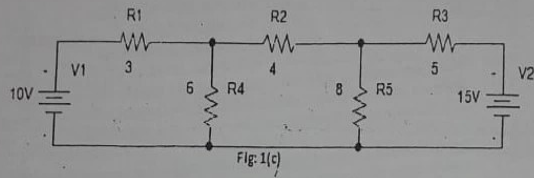


Fig: 1(c)

(d) Characterize the main properties of series and parallel circuit. 03

(a) State and explain KVL and KCL. 04

(b) Use nodal analysis to find various branch currents in the circuit of fig: 2(b) 04

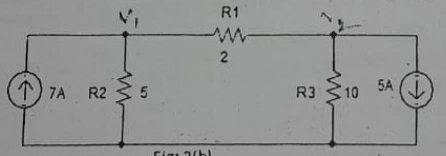


Fig: 2(b)

(c) Calculate the current through the 4Ω resistor of the circuit shown in fig: 2(c) by applying Thevenin's theorem. 04

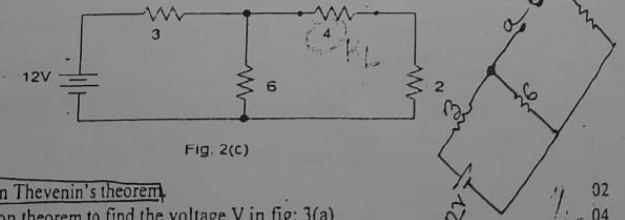


Fig. 2(c)

(d) State and explain Thevenin's theorem. 02

(a) Use Superposition theorem to find the voltage V in fig: 3(a) 04

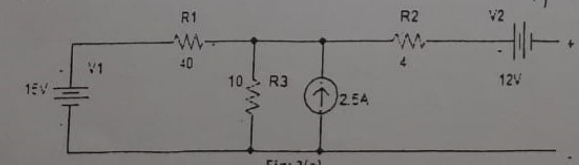


Fig: 3(a)

(b) Find current using Norton's theorem through a load of 8 Ω in the circuit shown in fig:3(b) 04

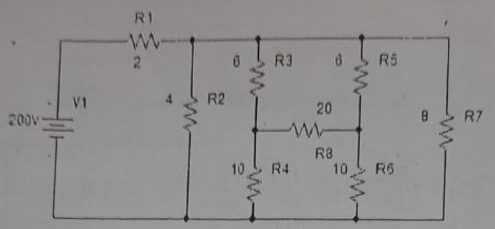


Fig: 3(b)

(c) Explain force between parallel conductors for currents in the both same and opposite direction. 04

(a) State and explain laws of magnetic force. 02

(a) Define magnetic circuit. Make a comparison between magnetic and electric circuit. 03

(b) Define magnetic hysteresis. Draw and describe hysteresis loop of a magnetic material. 04

(c) Write down the importance of hysteresis loop. 02

(d) Discuss Faradays laws of electromagnetic induction. 03

(e) Define self and mutual inductance. 02

(a) Discuss alternating current and voltage of an a.c circuit containing capacitance only. 03

(b) Define resonance in R-L-C series circuit. What are the effects of series resonance? Discuss the bandwidth of a series resonance circuit. 04

(c) A circuit having a resistance of 6Ω and inductive reactance of 8Ω is connected in parallel with another circuit having a resistance of 8Ω and capacitive reactance of 6Ω. The parallel circuit is connected across 200V, 50 Hz supply. Calculate: (i) supply current (ii) power factor of the whole circuit (iii) power consumed and (iv) the resistance and reactance of a series circuit which will take the same current at the same p.f. as the parallel circuit 04

(a) Define power factor. Discuss true and reactive power by using power triangle. 03

(a) Why effective value is used in alternating current? Determine R.M.S value of sinusoidal voltage. 04

(b) A capacitor of 50μF is connected in parallel with a coil that has a resistance of 20Ω and inductance of 0.05H. If this parallel combination is connected across 200V, 50Hz supply. Calculate (i) the line current (ii) power factor and (iii) power consumed. 04

(c) A pure inductance of 318mH is connected in series with a pure resistance of 75Ω. The circuit is supplied from 50 Hz source and the voltage across 75Ω resistor is found to be 150V. Calculate the supply voltage and the phase angle 03

(d) Calculate the capacitance of a condenser to be connected in series with a 100V, 80W lamp to enable it to be used on a 200V, 50Hz supply. What is p.f. of the circuit? 03

Patuakhali Science and Technology University

B.Sc Engg.(CSE) Level-1 Semester-I Final Examination: 2015 (January-June)

Course Code: EEE 111 Course Title: Basic Electrical Engineering

Credit Hour: 3.0 Full Marks: 70 Duration: 3 Hours.

[Figures in the right margin indicate full marks. Split answering of any question is not recommended.]

[Answer any 5 of the following questions. Resistance values are in ohms]

[1] a. Define resistance. What are the laws of resistance? 03

b. State and explain Ohm's law. 03

c. A 50-ohm resistor is in parallel with 100 ohm resistor. Current in 50 ohm resistor is 7.2 A. How will you add a third resistor and what will be its value of the line current if line current is to be 12.1 A? 02

d. Find the value of *R* and *Vs* in the fig:1(d). Also find the power supplied by the source. 04

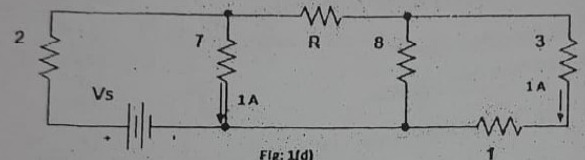


Fig: 1(d)

e. Discuss about short circuits with necessary diagrams for both series and parallel circuit. 02

[2] a. Define the following network terms: i) Bilateral Circuit ii) Parameters iii) Mesh iv) Active network 03

b. Write the impedance matrix of the network shown in figure 2(b) and find the current through 2Ω resistor. 05

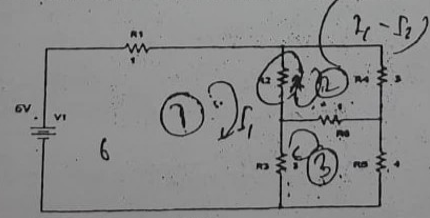


Fig: 2(b)

c. Calculate the current through the 2Ω resistor of the circuit shown in fig: 2(c) by applying Thevenin's theorem. 04

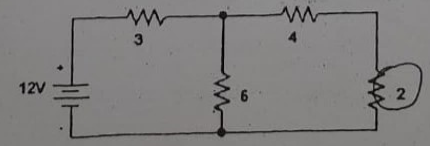


Fig: 2(c)

1. State and explain Norton's theorem.

[3] a. Define dependent source. Show different dependent sources with its circuit diagram. 03

b. Using Kirchhoff's current law, find the values of the currents i_1 and i_2 in the circuit shown in fig: 3(b). 03

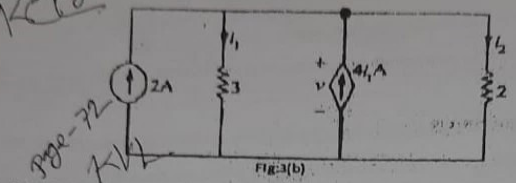


Fig:3(b)

c. Apply Kirchhoff's voltage law, to find the values of current i and the voltage drops V_1 and V_2 03
in the circuit shown in fig: 3(c). What is the voltage of the dependent source?

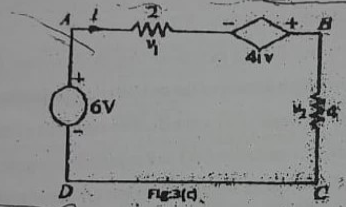


Fig:3(c)

d. Using superposition theorem, find the current through the 40 Ω resistor of the circuit shown in 05
fig: 3(d)

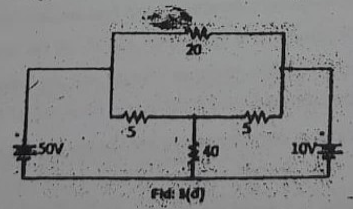


Fig: 3(d)

[4] a. Discuss right hand rule for the conductor and the coil. 03

b. Find the input resistance of the circuit between the points A and B of figure 4(b) 04

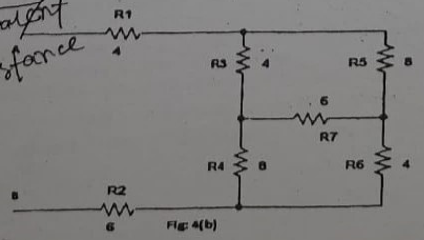


Fig: 4(b)

c. Explain the phenomena of force production between two parallel conductors for same and opposite current direction. 04

Show the comparison between magnetic and electric circuits. 03

State and explain Fleming's right hand rule for induced current. 02

Define resonance in R-L-C series circuit. What are the effects of series resonance? Discuss the bandwidth of a series resonance circuit. 05

What do you understand by effective value of an alternating current? Determine R.M.S value of sinusoidal current. 04

d. A pure inductive coil allows a current of 10A to flow from a 220V,50 Hz supply. Find (i) inductive reactance (ii) inductance of the coil (iii) power absorbed. 03

[6] a. A circuit having a resistance of 5Ω and inductive reactance of 8Ω is connected in parallel with another circuit having a resistance of 8Ω and capacitive reactance of 5Ω. The parallel circuit is connected across 200V, 50 Hz supply. Calculate: (i) supply current (ii) power factor of the whole circuit (iii) power consumed and (iv) the resistance and reactance of a series circuit which will take the same current at the same p.f. as the parallel circuit. 05

b. A 100 V, 60W lamp is to be operated on 220 V, 50 Hz supply. Find the value of (i) non-inductive resistance (ii) pure inductance in series with the lamp so that the lamp is not overrun. Which would be preferable? 04

c. Drive the equations of alternating current and voltage of an a.c circuit containing capacitance only. 03

d. The voltage applied to a parallel circuit is 68+j54 volts. The current in one branch is 10+ j14 amperes and the current in the other branch is 4+j8 amperes. What is the circuit power factor? 02

Patuakhali Science and Technology University

B.Sc.Engg.(CSE) Level-I Semester-I Final Examination-2018(January-June),Session:2017-2018

Course Code: PHY 111 Course Title: Physics-I

Credit Hour: 3.0 Full Marks: 70 Time: 3 hours

[Answer any 5(five) of the following questions. Numbers on the right margin indicate full marks.]

1. (a) Give the essential features of Brownian motion for the gaseous state of matter. 3
(b) State and prove the Maxwell's law of equipartition of energy. 4
(c) Explain the concept of degrees of freedom by taking suitable examples. 3
(d) Calculate the values of the molar heat capacities C_v and C_p of a gas, if the ratio of the heat capacities is 1.33. What is the atomicity of the gas? Given R=8.31 J/mol K 4

2. (a) Write short notes on:
(i) Zeroth law, (ii) Thermodynamic state and (iii) Thermodynamic equilibrium. 3
(b) Give some examples of Carnot Cycle in practice. Discuss the four process of a Carnot's reversible heat engine. 2+5
(c) Carnot engine working between a source temperature of T_2 and sink temperature of T_1 has efficiency of 25%. If the sink temperature is reduced by 20^0C, the efficiency is increased to 30%. Find the source and the sink temperature. 4

3. (a) What is Nernst's heat theorem? What do you infer form that? 3
(b) Assuming the concept of 'Heat Death of Universe'. 3
(c) Define entropy and disorder. How can we define entropy in terms of second law of thermodynamics? 2+2
(d) In an isothermal process at constant temperature, show that the change in entropy is $\Delta S_T = R\ln(V_2/V_1)$ of an ideal gas. 4

4. (a) What are Lissajous' figures? Calculate the resultant of two simple harmonic motions of equal time period when they act at right angles to one another and when the phase difference is zero and $\frac{\pi}{2}$. 1+3+3
(b) Two SHMs acting simultaneously on a particle are given by the equations $y_1=2\sin(\omega t + \pi/6)$ and $y_1=3\sin(\omega t + \pi/3)$. Calculate i) amplitude, ii) phase constant and iii) time period of the resultant vibration. 3
(c) Distinguish between damped and undamped vibrations and hence discuss the phenomenon of resonance. 4

5. (a) Explain simple harmonic motion and discuss its characteristics. 2+2
(b) Calculate the average kinetic energy and the total energy of a body executing simple harmonic motion. 6
(c) In one complete vibration for a vibrating particle show that the changes in displacement, velocity and acceleration for angles $0, \frac{\pi}{2}, \pi$ and $\frac{3\pi}{2}$. 4

6. (a) From the analytical discussion of formation of a stationary wave at an open end organ pipe, find out the following features of the resultant wave.
i) Displacement ii) Amplitude iii) Velocity iv) Acceleration v) Strain 5+7
Also, show the changes of the above features with respect to position, pointing the formation of nodes and anti-nodes.
(b) The equation of a transverse wave on a stretched string is, $y=A\sin 2\pi(vt-x-\emptyset)$. Write down the equation of a wave that would produce a stationary wave in the string on superposition with the given wave. 2

Patuakhali Science and Technology University

B.Sc.Engg (CSE) Level-I Semester-I Final Examination-2017(January-June),Session:2016-2017

Course Code: PHY 111 Course Title: Physics-I

Credit Hour: 3.0 Full Marks: 70 Time: 3 hours

[Answer any 5(five) of the following questions. Numbers on the right margin indicate full marks.]

1. (a) Give the essential features of Brownian motion for the gaseous state of matter. 3
(b) State and prove the Maxwell's law of equipartition of energy. 4
(c) Explain the concept of degrees of freedom by taking suitable examples. 3
(d) Obtain the atomicity for mono-,di- and triatomic molecules in the absence of vibratory motion. 4

2. (a) Write short notes on:
(i) Zeroth law, (ii) Thermodynamic state and (iii) Thermodynamic equilibrium. 3
(b) Explain why the specific heat at constant pressure C_p is greater than that C_v, the specific heat at constant volume. 3
(c) Give some examples of Carnot Cycle in practice. What is the principle used in the working of a refrigerator? 1+4
(d) Carnot engine working between a source temperature of T_2 and sink temperature of T_1 has efficiency of 25%. If the sink temperature is reduced by 20^0C, the efficiency is increased to 30%. Find the source and the sink temperature. 3

3. (a) What is Nernst's heat theorem? What do you infer form that? 3
(b) Assuming the concept of 'Heat Death of Universe'. 3
(c) Define entropy and disorder. How can we define entropy in terms of second law of thermodynamics? 2+2
(d) In an isochoric process at constant volume, show that the change in entropy is $\Delta S_v = C_v\ln(T_2/T_1)$ of an ideal gas. 4

4. (a) Distinguish between damped and undamped vibrations and hence discuss the phenomenon of resonance. 2+2
(b) Show that the average kinetic energy of a vibrating particle is given by $E = \pi^2 ma^2 n^2$. 3
(c) What are Lissajous' figures? Calculate the resultant of two simple harmonic motions of equal time period when they act at right angles to one another and when the phase difference is zero and $\frac{\pi}{2}$. 1+3+3

5. (a) Define phase velocity and wave velocity. Show that, for a single wave propagating in any medium, wave velocity = phase velocity. 5
(b) Calculate the energy density of a plane progressive wave. 9

6. (a) From the analytical discussion of formation of a stationary wave at an open end organ pipe, find out the following features of the resultant wave
i) Displacement ii) Amplitude iii) Velocity iv) Acceleration v) Strain 5+7
Also, show the changes of the above features with respect to position, pointing the formation of nodes and anti-nodes.
(b) The equation of a transverse wave on a stretched string is, $y = A\sin\frac{2\pi}{\lambda}(vt - x - \emptyset)$. Write down the equation of a wave that would produce a stationary wave in the string on superposition with the given wave. 2

Patuakhali Science and Technology University

B.Sc.Engg.(CSE) Level-1 Semester-1 Final Examination-2016(January-June),Session:2015-16

Course Code: PHY 111 Course Title: Physics-I

Credit Hour: 3.0 Full Marks: 70 Duration: 3 hours

[Figures in the right margin indicate full marks. Split answering of any question is not recommended.]

Answer any 5(five) of the following questions.

1.(a) Define (i) Law of Equipartition of energy and (ii) Brownian motion. 3+2

(b) Explain the term 'degrees of freedom'. How many the degrees of freedom will monoatomic, diatomic and triatomic molecules have in the absence of vibratory motion? 6

(c) Calculate the values of the molar heat capacities C_v and C_p of a gas, if the ratio of the heat capacities is 1.33.What is the atomicity of the gas? Given R=8.31J/ mol-k. 3

2.(a) Show that total change of entropy (i) for any reversible cyclic change and (ii) in any irreversible process. 5.

(b) In an isochoric process at constant volume, show that the change in entropy is $\Delta S_v = C_v \ln(T_2/T_1)$ of an ideal gas. 3

(c) 'Entropy is also called thermal inertia'-why? 2

(d) Calculate the change in entropy when 10grams of ice at 0^0C is converted into steam at 100^0C. 4

3.(a) Write short notes on:

(i) Thermodynamic state,(ii) Thermodynamic equilibrium,(iii) Carnot cycle and (iv) Nernst law. 1+1+3+2

(b) What is the principle used in the working of a refrigerator? 4

(c) Define coefficient of performance. Is it greater than 1?Explain. 3

4.(a) What is simple harmonic motion? Discuss its characteristics. 4

(b) Derive the differential equation of a simple harmonic motion. 3

(c) Show that energy of a plane progressive wave is given by $E = 2\pi^2 \rho n^2 a^2$. 7

5.(a) Explain Doppler effect. 2

(b) Obtain an expression for the apparent frequency of the note when the source

(i) moves towards the stationary observer,

(ii) moves away from the stationary observer. 3+3

(c) Write down the practical applications of Doppler effect. 3

(d) A person is standing near a railway track and a train moving with a speed of 36km/hr is approaching him. The apparent pitch of the whistle as heard by the person is 700hertz.Calculate the actual frequency of the whistle. Velocity of sound=350m/s. 3

6.(a) Prove that Coulomb's law is in accordance with Newton's third law of motion. 3

(b) State Gauss's law and express it in differential form and show that $\vec{\nabla}.\vec{E} = \rho/\varepsilon_0$. 1+4

(c) Coulomb's law is a special case of Gauss's law. Explain. 6

7.(a) Describe the Storing Energy in a Capacitor. 6

(b) For an electric dipole, show that the potential V of a dipole moment $\vec{P}$ and located at the origin is given by $\frac{1}{4\pi\varepsilon_0}\frac{\vec{P}.\vec{r}}{r^3} = -\frac{1}{4\pi\varepsilon_0}\vec{P}.\vec{\nabla}(\frac{1}{r})$. 8

Patuakhali Science and Technology University

B.Sc.Engg.(CSE) Level-1 Semester-1 Final Examination-2015(January-June),Session:2014-15

Course Code: PHY 111 Course Title: Physics-I

Credit Hour: 3.0 Full Marks: 70 Duration: 3 Hours.

[Figures in the right margin indicate full marks. Split answering of any question is not recommended.]

Answer any 5(five) of the following questions.

1.(a) Explain the term 'degrees of freedom' and the 'law of equipartition of energy' of a dynamical system. 3

(b) Show that for a gas possessing f degrees of freedom , the ratio of the two specific heats is $1+\frac{2}{f}$. Hence for monatomic and diatomic gas calculate the ratio of the two specific heats. 2+3

(c) In a vertical isothermal column for a Brownian particles, obtain the expression $n = n_0[e^{-mgx_2/kT} - e^{-mgx_1/kT}]$. 6

2.(a) Discuss the meaning and significance of the term 'heat death' of the universe. 4

(b) Calculate the entropy changes of an ideal gas and show (i)$\Delta S_T = R\ln(P_1/P_2)$,(ii) $\Delta S_P = C_p\ln(T_2/T_1)$ for the case of isothermal process and isobaric process respectively. 3+3

(c) Prove that the change in entropy for a reversible cycle is always zero. 4

3.(a) Discuss Carnot's reversible heat engine. Show that the total work done, $W = nRT_1(T_2 - T_1)\log_e\frac{V_b}{V_a}$ for a Carnot's reversible heat engine. 2+8

(b) Carnot engine working between a source temperature of T_2 and sink temperature of T_1 has efficiency of 20%. If the sink temperature is reduced by 10^0 C, the efficiency is increased to 25%. Find the source and the sink temperature. 4

4.(a) "The electric flux through any closed surface is proportional to the enclosed electric charge"-using this idea show that $\vec{\nabla}.\vec{E} = \frac{\rho}{\varepsilon_0}$. 4

(b) Given the electric field in a region of space $\vec{E} = 2x\hat{i} + 2y\hat{j} + 5z\hat{k}$.Calculate the volume charge density. 3

(c) Apply Gauss's theorem to calculate the electric field due to an infinitely long uniformly charged straight wire. 7

5.(a) What is Capacitor? Explain its charging and discharging process. 5

(b) Describe the Storing Energy in a Capacitor. 6

(c) A 2000 mF capacitor is charged through a 1 kW resistor using a 6 V supply. Calculate (i) the charging current after 2.5 second.(ii) the charge on the plates after 2.5 second. 3

6.(a) Explain –(i)free vibration,(ii)forced vibration,(iii)resonance. 3

(b)Show that in the case of forced vibrations, $m\frac{d^2y}{dt^2} + ky + \mu\frac{dy}{dt} = F\sin Pt$ .Hence explain the phenomenon of resonance. 6

(c) Derive the differential equation of damped oscillatory motion and give its general solution. 5

7.(a) Write down the common properties of an oscillating motion. 4

(b) Calculate the average kinetic energy and the total energy of a body executing simple harmonic motion. 6

(c) The motion of a particle in simple harmonic motion is given by $x = a\sin\omega t$. If it has a speed v when the displacement is x_1 and speed v' when the displacement is x_2 ,show that the amplitude of the motion is $a = \left[\frac{v^2x_1^2 - v'^2x_2^2}{v^2 - v'^2}\right]^{1/2}$. 4

Patuakhali Science and Technology University

Faculty of Computer Science and Engineering

1st Semester (L-1, S-I) Final Examination of B.Sc. Engg.(CSE) Jan- June-2018,Session:2017-18

Course Code: MAT-111 Course title: Mathematics-I

Credit Hour:3.0 Full marks:70 Time: 3.0 hours

[Figure in the right margin indicates full marks. Split answering of any question is not recommended]

Answer any 5 of the following questions

1. a) Define function with examples 2

b) Determine the domain and range of the following functions 3

$$f(x)=\begin{cases}3 & when \quad -3\le x<-1\\ -6x-3 & when \quad -1\le x\le 0\\ 3x-3 & when \quad 0<x\le 1\end{cases}$$

c) If $f(x)=\begin{cases}1+2x & when \quad -\frac{1}{2}\le x<0\\ 1-2x & when \quad 0\le x<\frac{1}{2}\\ -1+2x & when \quad x>\frac{1}{2}\end{cases}$, find the value of $\lim_{x\to\frac{1}{2}} f(x)$ 3

d) Test the continuity and differentiability of the function $f(x)=\begin{cases}x^2 & when \quad 0<x<1\\ x & when \quad 1\le x<2\\ \frac{x^3}{4} & when \quad 2\le x<3\end{cases}$ 6

At the point $x=2$

2. a) Find the differential coefficient of $\tan^{-1}\frac{\sqrt{1+x^2}-1}{x}$ with respect to $\tan^{-1}x$ 4

b) Derive the nth derivative of $y=\cos(ax+b)$ 4

c) State the Leibnitz's theorem 2

d) If $y=\sin(m\sin^{-1}x)$ then prove that $(1-x^2)y_{n+2}-(2n+1)xy_{n+1}+(m^2-n^2)y_n=0$ 4

3. a) If the straight line $x\cos\alpha+y\sin\alpha=p$ touches the curve $\left(\frac{x}{a}\right)^{\frac{n}{n-1}}+\left(\frac{y}{b}\right)^{\frac{n}{n-1}}=1$, then show that $(a\cos\alpha)^n+(b\sin\alpha)^n=p$ 5

b) State the generalized mean value theorem 2

c) Prove that $f(b)-f(a)=cf'(c)\ln\frac{b}{a}$ using the above theorem mention in (b). 2

d) If $x+y=2$, then determine the extreme values of $u=\frac{4}{x}+\frac{36}{y}$ 5

4. Integrate the following functions: 3.5 × 4

(i) $\int\frac{dx}{(1-x)\sqrt{1+x}}$ (ii) $\int\frac{dx}{(x+1)\sqrt{1+2x-x^2}}$ (iii) $\int\frac{dx}{(x^2+1)\sqrt{x^2+4}}$ (iv) $\int\frac{dx}{(x-1)\sqrt{x^2+1}}$

5. a) Write down the physical meaning of $\int_2^3 x^2\,dx$ 2

b) Find the value of $\int_0^{\frac{\pi}{2}}\frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}}dx$ 5

c) Establish the relation between Gamma and Beta function 4

d) Prove that $\Gamma\frac{1}{2}=\sqrt{\pi}$ 3

6. a) If $I_{m,n}=\int\cos^m x\cos nx\,dx$, then prove that $I_{m,n}=\frac{\cos^m x\sin nx}{m+n}+\frac{m}{m+n}I_{m-1,n-1}$. Also evaluate $\int_0^{\frac{\pi}{2}}\cos^5 x\cos 4x\,dx$ using this relation 6

b) Find the summation of the following series using definite integral.

$$\lim_{n\to\infty}\left[\frac{1}{n}+\frac{1}{\sqrt{n^2-1}}+\frac{1}{\sqrt{n^2-2^2}}+\ldots+\frac{1}{\sqrt{n^2-(n-1)^2}}\right]$$

c) Determine the area of the space enclosed by the curve $y^2=4ax$ and $x^2=4ay$

Patuakhali Science and Technology University

1st semester (L-I, S-I) Final Examination of B.Sc. in Engg. (CSE), Jan-June-2017

Session: 2016-17, Course Code: MAT-111, Course Title: Mathematics-I

Marks-70, Time: 3 hours, Credit: 3.00

[Figure in the right margin indicates full marks. Split answering of any question is not recommended]

Answer any 5 of the following questions.

1. a) Discuss about the following terms: 4

Function, Limit, Continuity and Differentiability of a function

b) Test the continuity and differentiability of the following function 6

$$f(x)=\begin{cases}x^2 & when \quad x\le 0\\ x & when \quad 0<x<1\\ \frac{1}{x} & when \quad x\ge 1\end{cases}$$

at the points $x=0$ and $x=1$

Find the differential coefficient of the function $\tan^{-1}\frac{2x}{1-x^2}$ with respect to $\sin^{-1}\frac{2x}{1+x^2}$ 4

2. a) Find the nth derivative of the function $y=\cos(ax+b)$ 5

b) State and prove the Leibnitz's theorem. 5

c) If $y=\tan^{-1}x$ then show that $(1+x^2)y_{n+2}+2(n+1)xy_{n+1}+n(n+1)y_n=0$ 4

3. a) State and prove the Roll's theorem. 5

b) Verify the Rolle's theorem for the following function: 5

$$f(x)=\begin{cases}x^2+1 & when \quad 0\le x\le 1\\ 3-x & when \quad 1<x\le 2\end{cases}$$

in the interval [0,2]

c) Determine the curvature of the curve $(x^2+y^2)^2=a^2(y^2-x^2)$ at the point (0,a) 4

4. Integrate the following functions: 14

(i) $\int\frac{x+2}{(x^2+3x+3)\sqrt{x+1}}dx$ (ii) $\int\frac{dx}{(1+x^2)\sqrt{1-x^2}}dx$ (iii) $\int\frac{2\sin x+\cos x+3}{\sin x+2\cos x+4}dx$

5. a) Find the value of $\int_0^{\frac{\pi}{2}}\log(\sin x)\,dx$ using definite integral properties 5

b) Prove that $\int_0^{\frac{\pi}{2}}\sin^m x\cos^n x\,dx=\frac{\Gamma\frac{m+1}{2}\Gamma\frac{n+1}{2}}{2\,\Gamma\frac{m+n+2}{2}}$ and hence evaluate $\int_0^{\frac{\pi}{6}}\cos^6 3x\sin^4 6x\,dx$ 6+3

6. a) If $I_{m,n}=\int\cos^m x\cos nx\,dx$ then show that $I_{m,n}=\frac{\cos^m x\sin nx}{m+n}+\frac{m}{m+n}I_{m-1,n-1}$ and from this relation determine the value of $\int_0^{\frac{\pi}{2}}\cos^4 x\cos 3x\,dx$ 6+3

b) Calculate the value of the series $\lim_{n\to\infty}\left[\frac{1}{n}+\frac{1}{\sqrt{n^2-1}}+\frac{1}{\sqrt{n^2-2^2}}+\ldots\ldots+\frac{1}{\sqrt{n^2-(n-1)^2}}\right]$ using definite integral. 5

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Patuakhali Science and Technology University

1st semester (L-1, S-I) Final Examination of B.Sc. in Engg. (CSE), Jan-June-2016

Session: 2015-16, Course Code: MAT-111, Course Title: Mathematics-I

Marks-70, Time: 3 hours, Credit: 3.00

[Figure in the right margin indicates full marks. Split answering of any question is not recommended.]

Answer any 5 of the following questions.

1 a) Define Limit of a function and Continuity. Test the continuity of the function 2+3=5

$$f(x)=\begin{cases}\frac{|x|}{x} & ,x\neq 0\\ 1 & ,x=0\end{cases}$$

at $x=0$.

b) Explain differential coefficient.

c) (i) Find the differential coefficient of $f(x)=x^{\cos^{-1}x}+(\sin x)^{\log x}$ with respect to x. 3x2=6 10

(ii) Differentiate $\tan^{-1}\frac{2x}{1-x^2}$ with respect to $\cos^{-1}\frac{1-x^2}{1+x^2}$

2 a) Define Partial Derivative 3

b) State and Prove Euler's theorem. 2+4=6 12

c) If $V=\tan^{-1}(x^2+y^2)/(x-y)$, then $xV_x+yV_y=\frac{1}{2}\sin 2V$ 5

3 a) State and Prove Leibnitz's theorem. 2+4=6 11

b) If $y=\sin(m\sin^{-1}x)$, then $(1-x^2)y_{n+2}-(2n+1)xy_{n+1}+(m^2-n^2)y_n=0$ 5

4 a) Find the equation of tangent and Normal at (x_1,y_1) to the curve $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ 3

5 Compute the integration of the following functions 3.5×4

(i) $\int\frac{dx}{\sqrt{(x-\alpha)(x-\beta)}}$ (ii) $\int\frac{x^2+2x+3}{\sqrt{x^2+x+1}}dx$

(iii) $\int\frac{dx}{x^2\sqrt{x+1}}$ (iv) $\int\frac{dx}{(x^2+1)\sqrt{x^2+4}}$ 10

6 a) Explain the physical meaning of definite integral and list four properties of definite integral 2+2 2

(5)

1. Rolle's theorem
2. Leibnitz's
3. Euler's theorem

Patuakhali Science and Technology University

B.Sc. Engg. (CSE) 1st Semester Level-1, Semester-I, Final Examination January-June/2018

Course code: CIT-111 Course Title: Programming Language

Credit hour: 3.00 Full marks: 70 Duration: 3 hours Session: 2017-18

[Figures in the right margin indicate full marks.]

Answer any 5 of the following questions. Split Answering is not recommended.

1 a. Draw the flowchart of adding 2 numbers if both are even. 4

b. Compare the memory requirements of different data types. 3

c. Show the validity of the following variable names: 1x7
(i) int, (ii) Double, (iii) 5C, (iv) __, (v) _5C, (vi) 1_ab, (vii) xyz

2 a. Explain different types of programming errors with examples. 4

b. Find the errors, if any, in the program at the right hand side. Mention the type of each error and fix it. 10

```
Float a, b, Int c = 0;
scafn("%d%d" &a, c);
a = ++b++ - c++;
b = a%c_
If(a<b,c)
    pritnf("%d%d%d",a,b,c);
```

3 a. What will be the values of *a* and *b* after executing following code segment? 2

```
int a, b = 10;a = -b--;
```

b. What will be the output to the following program? 2

```
char *getString()
{   char *str = "test for strings";
   return str;
}
```

```
int main()
{  printf("%s", getString());
    return 0;
}
```

c. What will be the output to the following program? 3

```
int main()
{
    int x;
    printf("%d",scanf("%d",&x));
/* Suppose that input value given for above scanf is 20 */
    return 1;       }
```

d. What will the following 2 printf statement print? 3

```
printf("size=%d ",sizeof("PSTU"));
```

```
printf("length=%d ", strlen("PSTU"));
```

e. What will the following code snippet print? 2

```
int first = 50, second = 60, third;
third = first /* Will this comment work? */ + second;
printf("%d /* And this? */ \n", third);
```

f. What will be the output of the following code snippet for *a* = 5 and *b* = 2? 2

```
if(a&b)    printf("inside if");
else   printf("inside else");
```

4. Develop a program in C to create a student database using file and structure. The structure will contain three strings named *firstName*, *lastName* and *studentID* of length maximum 20. Your program should have functions for read the file, write in the file and search in the file. Your program should at first print a menu as follows.

Press 1 for add new entry
Press 2 for show list
Press 3 to search by ID
Press 4 to search by last name
Your program should be able to perform the tasks mentioned in the menu.

2 marks for read file, 2 marks for write file, 2 marks for the menu, 2 marks for the structure.
2 marks for searching in the list, 2 marks for the main function. 2 marks for getting everything right.

5. a. Suppose you have the following program. 1 + 2 + 2

```
# include <stdio.h>
void fun(int x)
{
    x = 30;
}
```

```
void main()
{
  int y = 20;
  fun(y);  printf " d", y);
}
```

Will this program run? If yes, what will be the output? Justify your answer.

b. Compare *call by value* and *call by reference* with necessary examples. 4
c. Create a C program to swap 2 integer values using functions and pointers. 5

6. a. Compare the file openning modes – read, write and append. 2
b. Describe the functionality of *fopen*, *fseek*, *fprintf*, *fscanf*, *fgets*, *fputs* and *fclose* functions. 1x7
c. Suppose you do not have the *string.h* header file. You have to copy a string *str1* to another string *str2*. You also need to calculate the length of *str1*. Create a C program to copy *str1* into *str2* and to get the length of *str1*. Write 2 functions namely *void copyString(char *str1, char*str2)* and *int geStringtLength(char *str1)* and call them from the main function. Print the length of the string in the main function. 5

2 marks for *copyString* function, 1 mark for *getStringLength* function,
1 mark for the main function, 1 mark for getting everything right.

```
#include <stdio.h>
int main()
{
int a, b, i;
scanf("%d %d", &a, &b);
printf("Enter the value");
for(i=2, i<=n/2; i++)
```

Patuakhali Science and Technology University

B.Sc. Engg. (CSE) Level-1, Semester-I, Final Examination Jan-Jun/2017, Session: 2016-17
Course code: CIT-111 Course Title: Programming Language
Credit hours: 3.00 Full marks: 70 Duration: 3 hours

[Figures in the right margin indicate full marks.]
Answer any 7 of the following questions. Split answering is not recommended.

1. a. Draw the flow chart of solving a complex problem using modular programming. Assume your modules as smaller sub-problems. 3

b. Show the validity of the following variable names: 1x7
(i) int, (ii) Double, (iii) 5C, (iv) __ , (v) _5C, (vi) 1_ab, (vii) xyz

2. a. A program crashed after taking some inputs. What type of error had your program? 2

b. Find the errors, if any in the following program. Mention the type of each error and fix it.

```
int a, b; float c = 0;
scafn("%d%d", &a,&c);
a = ++b++ - c++;
b = a%c
fi(a<b,c)
    pritnf("%d%d%d",a,b,c);
```

3. a. Write a C program which should take 3 strings *str1*, *str2*, *str3* as inputs. Then, it should search *str2* within *str1* and replace all occurrences of *str2* with *str3*. 6

b. Write the syntax of ***for***, ***while*** and ***do-while*** loops. When should which one be used and why? 2+2

4. What will be values of a, b and c after executing the following code segments? 2.5x4

a)
```
int a = 5, b = 3, c, i;
for(i=0;i<10;i+=3);
{
    c = a+b*i;
    a = b%i;
    b = c-i;
}
```

b)
```
int a=5, b=6, c=1;
a = ++a + b-- * --c;
if(a&&b||c)
    a = a| c;
else
    a = b-c;
```

c)
```
int i=10, a=5, b=2, c=7;
do
{
    c -= a&b;
    a -= b--;
    i -= 2;
}while(i>0);
```

d)
```
int a=0,b=5,c=4;
a *= a++ + b-- + --c;
```

5. a. Write a C program to reverse a given integer. Your program should have a function named ***numReverse*** which should take an integer ***n*** as input and return the reversed number as output. Follow the sequence below. 6

Main function (2 marks)	***numReverse*** function (3 marks)
{	{
Input ***n***;	Reverse ***n***
Call ***numReverse*** function with ***n***	Return reversed ***n***
Print reversed ***n***	}
}	1 mark for getting everything right.

b. Write a C program to reverse a string. This program should contain a function named ***strReverse*** function which takes a string as input and returns the reversed string. The sequence should be similar to the one given in 5.a. with the substitution of the number with a string. 2+2

6. Write a program in C to create a telephone book using file and structure. The structure should contain three strings named ***firstName***, ***lastName*** and ***phNo*** of length maximum 20. Your program should have functions for read the file, write in the file and search in the file. Your program should at first print a menu as follows. 10

Press 1 for Search the phone book
Press 2 for save a new number
Press 3 to Exit
Your program should be able to perform the tasks mentioned in the menu.

2 marks for read file, 2 marks for write file, 2 marks for the menu, 2 marks for structure, 2 marks for main function

7. a. Compare the file openning modes – read, write and append. 3
   b. Describe the functionality of ***fopen***, ***fseek***, ***fprintf***, ***fscanf***, ***fgets***, ***fputs*** and ***fclose*** functions. 1x7

8. a. Define: pointer, array and string 1x3
   b. Suppose you declared int a[10], *p; 3
   Is it possible to write p = a; ? Justify your answer
   c. Compare array and structure. 2
   d. You want to take 5 students' name and their IDs as input from the keyboard. Declare the variables to perform this task. 2

9. a. Write a C program to swap 2 integer values using functions and pointers. 4
   b. What would have happened if you did not use pointer to pass the values in the function? Explain. 3
   c. Draw a flowchart for the process of compiling and running a C program. 3

Patuakhali Science and Technology University

B.Sc. Engg. (CSE) Level-I, Semester-I, Final Examination Jan-Jun/16, Session: 2015-16

Course code: CIT-111 Course Title: Programming Language

Credit hours: 3.00 Full marks: 70 Duration: 3 hours

[Figures in the right margin indicate full marks.]

Answer any 7 of the following questions. Split answering is not recommended.

1. a. Explain how a computer executes a given set of instruction? 2
   b. Show the validity of the following variable names: 1x7
   (i) int1, (ii) Int, (iii) 1ab, (iv) __, (v) _1ab, (vi) 1_ab, (vii) ab
   c. Calculate memory requirements for the declaration: int a=5,b=0; int c[500]; char s[100]; 1

2. a. A program crashed after taking some inputs. What type of error had your program? 2
   b. Provide appropriate examples of programming errors. 3
   c. Find the errors, if any in the following program. Mention the type of each error and fix it. 5

```
int a, b; float c = 0;
scanf("%d%d", &a, b);
a =* b;
```

3. a. What happens if you run the following 2 code segmentsfor the input given? Which one is better in terms of error handling? 1+2

```
int main()
{
charstr[10];
gets(str);
puts(str);
return 0;
}
```

a)

```
int main()
{  charstr[10];
int i;
 for(I =0; i<30; i++)
    scanf("%c",&str[i]);
 puts(str);
return 0; }
```

b)

Given Input:

I told him that I saw a saw to saw a saw with a saw.

   b. How can you find the nth odd number using a C program? 5
   c. Write the syntax of ***for***, ***while*** and ***do-while*** loops. When should which one be used and why? 1+1

4. What will be values of a, b and c after executing the following code segments? 2.5x4

a)
```
int a = 1, b = 1, c, i;
for(i=0;I<10;i++);
{
  c = a+b;
  a = b;
  b = c;
}
```

b)
```
int a=3,b=5,c=4;
a *= ++a + b-- + --c;
if(a&&b||c)
a %= c;
else
    a = b-c;
```

c)
```
int i=10, a=25, b=2, c=7;
do
{
   c -= a&&b;
   a-=b--;
   I--;
}while(i>0||a>0);
```

d)
```
int a=0,b=5,c=4;
a *= a++ + b-- + --c;
```

5. a. Write a C program to calculate the factorial of a number, *n*. Your program should contain a function named *fact* which should take the number, n as input and return the factorial value as output. Assume *0<= n <= 20*. Hint: 0! = 1. You may skip the preprocessors. 6

b. Number the lines of your program in answer to the question 5.a. Then write the sequence of lines for the execution of your program. 2

c. How did you pass the parameter in the *fact* function? Explain in terms of memory. 2

6. Write a C program to print the series and the result of the series $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \frac{x^n}{n!}$ 1.5x4 + 4

Your program should contain 4 functions named *input, output, calcFact* and *calcE*. Define the functions using the following prototypes.

```
void input(int *n, int *x);
long calcFact(int n);
double calcE(int n, int x);
void output(double e);
```

input function should take two pointer variables as parameter and these values will be taken from the keyboard. calcFact function should calculate the factorial of *n*. calcE should calculate the result of the series for *x* and *n*. output function should print the values of e.

7. a. Explain the read, write and append file opening modes when you open your file in text mode. Which mode is better if you want to add data time to time and why? 2+1

b. Suppose you have some integer numbers in a file named "*input.text*". You have to read the numbers and save the even numbers in "*even.txt*" file and the odd numbers in "*odd.txt*" file. Write a C program with functions named *input* and *output* for taking input from the file and saving the numbers into the file repectively. Assume the files are in the same directory where your program is located. 7

8. a. For a file pointer *fp*, exlpain what fp=fopen(.....) and feof(fp) will do. 1+1

b. Write and explain the syntax of *strcpy* and *strcat* functions for two strings named *str1* and *str2*. 1+1

c. Suppose, you are forbidden to use the strcpy and strcat functions. Therefore, you have to write them on your own. Now write two functions in C to perform these two operations. 3+3

9. a. Declare a structure in C for a teacher of CSE, PSTU. The structure should contain the name, designation and department of the teacher. Then, write a C program where you use this structure to take input from the keyboard and show output to the console. 4

b. Write a function in C using the following function prototype for swapping two values. 2+2

```
void swap(int *x, int *y);
```

Now call this function from the main function to swap two integer varaibles.

c. Describe what you have done in the main function while calling the swap function. Also, explain what happens in the swap function. 1+1

Patuakhali Science and Technology University

B.Sc.Engg.(CSE) Level-1 Semester-I, Final Examination-2018(January-June)

Course Code: CHE 111 Course Title: Chemistry

Credit Hour: 3.0 Full Marks: 70 Duration: 3 Hours

[Figures in the right margin indicate full marks. Split answering of any question is not recommended.]

Answer any 07(seven) of the following questions.

1. Define chemistry. Describe the branches of chemistry. What are the types of Quantitative analysis? Enlist the needs of analysis. 2+3+3+2 = 10

2. What do you mean by reversible and irreversible reaction? Give the characteristics of chemical equilibrium. Explain the mathematical expression of law of mass action. 3+3+4 = 10

3. How can you explain the term P^H and P^{OH}? Calculate the P^H and P^{OH} of pure water. Calculate the P^H of 0.001M HCl. 2+6+2 = 10

4. Define buffer solution with its types. Calculate the P^H of an acidic buffer solution. The components of acetic acid and sodium acetate in a buffer solution is 0.1 M. Calculate the pH of buffer solution, where pKa=4.73 3+5+2 = 10

5. What is meant by volumetric analysis? Write down the advantages and disadvantages of volumetric analysis. Explain the terms oxidation and reduction. 2+4+4 = 10

6. What is an indicator? Mention the name of indicators giving some information. Describe the two ionic theory of acid base indicator. 1+3+6 = 10

7. Clarify the different concept of acid and base. How can you prepare 0.1N Sulfuric acid and 0.5 N Sodium Carbonate Solution. 5+3+2 = 10

8. What are electrolytes? Prove that the degree of dissociation for a molar solution of weak electrolytes is equal to the square root of its dissociation constant. At 25^0C in a solution of 0.1 N of formic acid, 2.25 % of its dissociate. Calculate its dissociation constant. 2+6+2 = 10

9. Write a short notes on a) Basic ways of expressing concentration and b) Matter 5+5 = 10

Patuakhali Science and Technology University
B.Sc.Eng.(CSE) Level-1 Semester-I, Final Examination-2017(January-June)
Course Code: CHE 111 Course Title: Chemistry
Credit Hour: 3.0 Full Marks: 70 Duration: 3 Hours

[Figures in the right margin indicate full marks. Split answering of any question is not recommended.]

Answer any 07(seven) of the following questions.

1. Define chemistry. Describe the branches of chemistry. What are the types of Quantitative analysis? Enlist the needs of analysis. 2+3+3+2

2. What do you mean by reversible and irreversible reaction? Give the characteristics of chemical equilibrium. Explain the mathematical expression of law of mass action. 3+3+4

3. How can you explain the term P^H ? Calculate the P^H and P^{OH} of pure water. Calculate the P^H of 0.001M HCl. 2+6+2

4. Define buffer solution with its types. Calculate the P^H of an acidic buffer solution. The components of acetic acid and sodium acetate in a buffer solution is 0.1 M. Calculate the pH of buffer solution, where pKa=4.73. 3+5+2

5. What is meant by volumetric analysis? Write down the advantages of volumetric analysis. Show the partial equations of standardization of $KMnO_4$ by Mohr's salt solution. 2+2+6

6. What is an indicator? Mention the name of indicators giving some information. Describe the two ionic theory of acid base indicator. 1+3+6

7. Clarify the different concept of acid and base. 22.5 ml. of 0.102(N) $KMnO_4$ is required to oxidized 10 ml. Mohr's salt solution, determine the amount of Fe^{++} of that solution. 5+5

8. What are electrolytes? Prove that the degree of dissociation for a molar solution of weak electrolytes is equal to the square root of its dissociation constant. At 25^0C in a solution of 0.1N of formic acid, 2.25 % of it's dissociates. Calculate it's dissociation constant. 2+6+2

9. Write short notes on
   a) Basic ways of expressing concentration
   b) Matter 5+5

Patuakhali Science and Technology University
B.Sc.Eng.(CSE) Level-1 Semester-I, Final Examination-2016(January-June)
Course Code: CHE111 Course Title: Chemistry
Credit Hour: 3.0 Full Marks: 70 Duration: 3 Hours

[Figures in the right margin indicate full marks. Split answering of any question is not recommended.]

Answer any 07 of the following questions.

1. Define chemistry. Describe the branch of chemistry. What are the types of analysis? 2+3+5

2. What is law of mass action and equilibrium constant? Explain the mathematical expression of law of mass action. 3+7

3. How can you explain the term P^H and P^{OH}? Calculate the pH and pOH of pure water. Calculate the P^H of 0.001M Hcl. $pH = -\log_{10}[0.001] = 3$, 2+6+2

4. Define buffer solution, buffer action and buffer capacity. Describe the composition of buffer solution. Calculate the P^H of an acidic buffer solution. 3+2+5

5. What is meant by volumetric analysis? Write down the disadvantage and disadvantage of volumetric analysis. Show the partial equation of standardization of $KMnO_4$ by Oxalic acid. 1+3+6

6. Explain the term a) % solution b) equivalent weight c) normality and molality d) standardization. 10

7. Define oxidation and reduction on the basis of electronic concept and charge concept. Clarify the different concept of acid and base. 5+5

8. What are electrolytes? Prove that the degree of dissociation for a molar solution of weak electrolytes is equal to the square root of its dissociation constant. 2+8

9. Write a short notes on a) Solution b) Primary standard substance c) Action of basic buffer. 6+2+2

Patuakhali Science and Technology University
B.Sc.in CSE (Hons.) Level-1 Semester-I, Final Examination-2015 (January June)
Course Code: CHE111 Course Title: Chemistry
Credit Hour: 3.0 Full Marks: 70 Duration: 3 Hours

[Figures in the right margin indicate full marks. Split answering of any question is not recommended.]

Answer any 07 of the following questions.

1. Define chemistry And analysis. Describe the branch of chemistry. What are the types of analysis? 3+4+3

2. What is chemical equilibrium? Explain the mathematical expression of law of mass action. 3+7

3. How can you explain the term P^H and P^{OH} and establish a relationship between them? Calculate the P^H of 0.001M Hcl. 2+6+2

4. Define buffer solution with its types. How act buffer? Calculate the P^H of an acidic buffer solution. 3+3+4

5. What is meant by volumetric analysis? Write down the advantage of volumetric analysis. Show the partial equation of standardization of $KMnO_4$ by Oxalic acid. 2+2+6

6. Explain the term a) ppm b) equivalent weight c) Normality and Molarity d) Standardization. 10

7. Define oxidation and reduction on the basis of charge concept. Write down the rules for redox equation. Clarify the different concept of acid and base. 2+4+4

8. What are electrolytes? Describe the term common ion effect. Prove that the degree of dissociation for a molar solution of weak electrolytes is equal to the square root of its dissociation constant. 2+2+6

9. Write a short notes on a) Chemical bond b) Kp and Kc c) Matter d) P^H scale 10

Patuakhali Science And Technology University
Faculty of CSE
Final Exam Semester: 1st Batch : 14
Course Code: CCE 112 Course Title : Engineering Drawing
Total Marks 70 Time: 2.30 Hours Date : 30 December.2010

1. Typing Master work 10
2. Viva Voce 20
3. Write your cv using MS word where you use almost features of MS word 10
4. Make a presentation (10 slide) of your CV with MS power point features 10
5. Make MS Excel sheet with below table and questions 20

Employee Name	Department	Place	Basic	DA	TA	MA	HRA	Gross	PF	Tax	Net Salary	Designation
Borsha	Accounts	Barisal	9000									
Shorot	Purchase	Dhaka	12000									
Hemonto	Marketing	Patuakhali	6500									
Bosonto	Sale	Barisal	10000									
Rodela	Accounts	Barisal	15000									
Ekush	Sale	Patuakhali	7500									
Bijoy	Marketing	Dhaka	10500									
Nodee	Purchase	Patuakhali	8000									
Somudro	Accounts	Dhaka	13000									
Deeghi	Markering	Dhaka	8500									

i. Calculate DA @50% of basic to those employees whose department is "Sale" and who belong to "Barisal", others will get @40% of basic.
ii. Calculate TA @20% of basic to those employees whose department is "Marketing" or whose basic is more than 9000, others will get "0" (ZERO).
iii. Calculate MA @10% of basic to those employees whose basic is more than 12000; @5% of basic to those employees whose basic is more than 8000 but less than 12000, BDT: 200 whose basic is less than 8000.
iv. Calculate HRA @5% of basic to all employees.
v. Calculate Gross.
vi. Calculate PF fixed BDT: 1000 to those employees whose basic >= 6000 and department is "Accounts" and who belongs to "Dhaka", others will get BDT: 300.
vii. Calculate Tax @3% of gross to those employees whose basic is less than 9000 or whose department "Purchase", others will get @8% of gross.
viii. Calculate Net Salary.
ix. Give "Designation" as a "Officer" whose basic >= 12000, as a "Manager" whose basic >= 8000 but < 12000, as a "Clerk" whose basic < 8000.

* Make a folder in your computer of D drive as your name. Question 3, 4 and 5 , file name save as your id name.

Duration: 1hr 10mins Full marks: 40

1 Look before you Leap!! 10

You have to find out if a given year is a leap year.

Input

You will be given a year, y ($0<y\leq 12000$) which you'll need to check if y is a leap year.

Output

For each input y, you need to print "Leap" if y is a leap year, "Took" otherwise. For a negative input, you have to print "Sorry".

Sample Input

2000
1999
1996
2100
-1

Sample Output

Leap
Took
Leap
Took
Sorry

include <stdio.h>
int main

2 Bizarre Calculator 15

You have to make a calculator which adds, subtracts, multiplies and divides 3 numbers.

Input

You will be given 2 operators in the first line. The second line will contain 3 numbers. You have to perform the exact operations as the operators indicate if the summation of the first 2 numbers are greater than the last number. Otherwise, you have to perform a swapped operation on the numbers. For example, if the inputs are

+ /
6 3 12

Then you have to perform 6/3 + 12 as (6+3) is less than 12.

Output

You have to print the result.

Sample Input

+ /
6 3 12
+ /
12 6 3

Sample Output

14
6

3 Caesar Cypher 15

You have to encrypt a plain text to a cypher text using Caesar cypher technique. The technique is to move each character to a fixed length so that the plain text becomes unreadable. If any movement overcomes the last character, it automatically starts from the first character.

Input

The first line of input will contain an integer k ($0<k<10$).

The second line will have a string as a plain text. You'll have to move each character to k positions. If you go beyond "Z" or "z", you have to calculate from "A" or "a".

Output

Print one line with the cypher text

Sample Input

3
saw a saw to saw a saw

A. Answer the following questions. All question carry equal mark (5*3=15).

1. Differentiate between data and information.
2. Write down some advantages and disadvantages of using computer.
3. How the elements of an information system in an enterprise might interact?
4. Write short note on network, server and Internet.
5. What are the elements of an information system? Describe.

B. Fill in the gaps. All question carry equal mark (5*1=5).

1. Case that contains the electronic components of the computer that are used to process data is called _____.
2. _____ enables a computer to send and receive data, instructions, and information to and from one or more computers or mobile devices.
3. _____ involves reducing the electricity consumed and environmental waste generated when using a computer.
4. A _____ is a mobile computing device designed for single-player or multiplayer video games.
5. Fastest supercomputers are capable of processing more than one _____ instructions in a single second.